Lie Algebra

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1 Lie Algebra

I Matrix Lie group

A matrix Lie group over F is a closed subgroup of a general linear group over F.

II Lie group

A Lie group is a topological group $(X, \tau, *)$ such that (X, τ) is a finite-dimensional smooth manifold, the maps of $(x, y) \in X^2 \mapsto x * y$ and $x \mapsto x^{-1}$ are smooth maps.

III Lie Algebra

A Lie algebra is a vector space $\mathfrak{g}$ over a field F together with a binary operation $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ called the Lie bracket, satisfying the following axioms:

- Bilinearity:
$$[ax + by, z] = a[x, y] + b[y, z]$$
$$[z, ax + by] = a[z, x] + b[z, y]$$
for all $a, b \in F$ and $x, y, z \in \mathfrak{g}$.
- Alternating property:
$$[x, x] = 0$$
for all $x \in \mathfrak{g}$.
- Jacobi identity:
$$[x, [y, z]] + [z, [x, y]] + [y, [z, x]] = 0$$
for all $x, y, z \in \mathfrak{g}$.

IV Lie algebra of a Lie group

A Lie algebra of a Lie group is the tangent space at the identity.

V Lie's third theorem

Every finite-dimensional real Lie algebra is the Lie algebra of some simply connected Lie group.

Proof. TODO □

i (Lie) exponential map

Let G be a Lie group and $\mathfrak{g}$ be the Lie algebra of G. The exponential of $X \in \mathfrak{g}$ is defined as

$$\exp(X) = \gamma(1)$$

where $\gamma : \mathbb{R} \to G$ is the unique one-parameter subgroup of G whose tangent vector at the identity is equal to X.